KASNEB PASTPAPERS · CPA STUDY MATERIALS

Introduction to Computing

Course: DICT — Diploma in Information Communication Technology

Level: Level I

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Syllabus version: Based on the standard KASNEB DICT Level I Introduction to Computing syllabus topics — practical computer literacy, parts of a computer system, common software categories, basic file management, safe computing practices, and everyday Internet use. Authored from general professional ICT knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

> **Original work notice.** Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The worked example and practice questions are original scenarios.

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1. What is a computer system?

A **computer system** is the combination of hardware (the physical machine you can touch), software (the instructions that tell the hardware what to do), and data (the information being processed) working together to perform useful tasks. At this practical, entry level, the goal is confident everyday use of a computer — knowing its parts, its common software, and how to use and look after it safely — rather than deep technical theory.

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2. The parts of a desktop/laptop computer

- **System unit/case** — houses the core components: motherboard, CPU, memory, storage, and power supply.
- **Motherboard** — the main circuit board that all other components plug into or connect through.
- **Monitor** — the primary visual output device.
- **Keyboard and mouse/trackpad** — the primary input devices for typing and pointing.
- **Speakers/headphones** — audio output devices.
- **Webcam and microphone** — common input devices for video calls and audio recording.

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3. Categories of everyday software

- **Operating system** — the foundational software that manages the computer's hardware and provides the environment other programs run in (e.g. Windows, macOS, a Linux distribution).
- **Productivity software** — word processors, spreadsheets, and presentation programs used for everyday office tasks.
- **Web browsers** — software used to access and view websites on the Internet.
- **Communication software** — email clients, video-calling apps, and instant-messaging apps.
- **Utility software** — small programs that help maintain or optimise the computer, e.g. antivirus software, disk clean-up tools, and file compression tools.

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4. Files, folders, and basic file management

- A **file** is a named unit of stored data (a document, an image, a spreadsheet, and so on), usually identified by a name and an extension (e.g. `report.docx`, `photo.jpg`) that indicates its type.
- A **folder (directory)** is a container used to organise related files together, and folders can be nested inside other folders to build a clear hierarchy.
- Good file-management habits: use clear, descriptive file names; organise files into a sensible folder structure by topic or date; avoid duplicate copies scattered across the computer; regularly back up important files to a second location.

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5. Storage devices and units of measurement

- **Hard Disk Drive (HDD)** — traditional magnetic storage, generally offering large capacity at lower cost.
- **Solid State Drive (SSD)** — storage using flash memory with no moving parts, generally much faster than an HDD.
- **USB flash drive/external hard drive** — portable storage used to move files between computers or keep an extra backup.
- **Optical media (CD/DVD)** — older removable storage format, now much less commonly used.

Units of measurement, from smallest to largest, each roughly 1,000 times the previous one: byte → kilobyte (KB) → megabyte (MB) → gigabyte (GB) → terabyte (TB).

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6. Ports and connecting peripherals

- **USB (Universal Serial Bus)** — the most common general-purpose port, used to connect flash drives, keyboards, mice, printers, and many other peripherals.
- **HDMI** — used to connect a computer to an external monitor, projector, or television, carrying both video and audio.
- **Ethernet port** — used to connect a computer to a wired local area network.
- **Audio jack** — used to connect wired headphones, speakers, or a microphone.

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7. Safe computing practices

- Use a strong, unique password (or passphrase) for each account, and enable multi-factor authentication where it is offered.

- Keep the operating system and installed applications updated with the latest security patches.
- Install and maintain a reputable antivirus/anti-malware program, and keep it updated.
- Avoid opening email attachments or clicking links from unknown or suspicious senders.
- Lock the computer (or log off) whenever stepping away from it in a shared space.
- Keep regular backups of important files, stored on a separate device or service from the main computer.

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8. Basic troubleshooting

A simple, logical approach to common everyday problems:

1. **Restart the device** — resolves a large share of minor glitches by clearing temporary faults.
2. **Check physical connections** — for a peripheral that isn't working, confirm cables/plugs are properly connected.
3. **Check for updates** — outdated software or drivers are a common cause of malfunctioning hardware or applications.
4. **Free up storage space** — a nearly full storage drive can slow a computer down or cause application errors.
5. **Consult the error message** — read what the computer is actually telling you before assuming the worst; many messages point directly at the cause.
6. **Escalate if unresolved** — if simple steps don't fix it, seek qualified technical support rather than attempting risky fixes (e.g. inside the power supply) without training.

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9. Using the Internet safely and effectively

- A **web browser** is used to access websites, identified by their web address (URL); a **search engine** helps locate information across the web using keywords.
- Look for signs a website is likely legitimate/secure (e.g. a valid padlock/HTTPS indicator in the browser) before entering sensitive information such as passwords or payment details.
- Be cautious of "phishing" attempts — messages or fake websites designed to trick you into revealing passwords or personal/financial information.
- Practise good digital etiquette ("netiquette") in emails and messaging: be clear, courteous, and mindful of tone, since text lacks the non-verbal cues of face-to-face communication.

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10. Computer ergonomics and care

- Position the monitor at roughly eye level and an arm's length away, to reduce neck and eye strain.
- Sit with feet flat on the floor and good back support, to reduce posture-related strain over long periods of use.
- Take short, regular breaks from the screen to rest your eyes and stretch.
- Keep the computer in a clean, dust-free, well-ventilated location, and avoid eating or drinking directly over the keyboard.

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11. Full worked question — storage capacity calculation

Scenario. A student wants to store photos on a 64 GB USB flash drive. Each photo file is approximately 4 MB in size. The flash drive already has 8 GB of other files stored on it.

Required: Estimate how many more photos can be stored on the flash drive, showing your workings. (Assume 1 GB = 1,000 MB for simplicity.)

Solution:

Total capacity = 64 GB = 64 × 1,000 MB = 64,000 MB

Space already used = 8 GB = 8 × 1,000 MB = 8,000 MB

Remaining free space = 64,000 − 8,000 = 55,000 MB

Number of additional photos that fit = Remaining free space ÷ size per photo

= 55,000 MB ÷ 4 MB

= **13,750 photos** (approximately)

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12. Practice examination — KASNEB-style paper (original, not a past paper)

> **Note on format.** This section is laid out the way a KASNEB DICT examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

DICT LEVEL I — INTRODUCTION TO COMPUTING (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

(a) Define a computer system, identifying its three main components. *(6 marks)*

(b) Describe FOUR physical parts of a desktop or laptop computer, stating the function of each. *(8 marks)*

(c) Distinguish between an operating system and productivity software, giving one example of each. *(6 marks)*

(Total: 20 marks)

QUESTION TWO

(a) Explain THREE good file-management habits that help keep a computer's files organised. *(9 marks)*

(b) Distinguish between a Hard Disk Drive (HDD) and a Solid State Drive (SSD). *(6 marks)*

(c) Arrange the following storage units from smallest to largest: gigabyte, byte, terabyte, megabyte, kilobyte. *(5 marks)*

(Total: 20 marks)

QUESTION THREE

A user wants to copy files onto a 32 GB flash drive that already has 5 GB of files stored on it. Each file to be copied is approximately 250 MB in size.

Required: (Assume 1 GB = 1,000 MB.)

(a) Compute the remaining free space on the flash drive, in MB. *(6 marks)*

(b) Compute how many of the 250 MB files can be copied onto the remaining free space. *(8 marks)*

(c) State TWO other types of portable/removable storage besides a USB flash drive. *(6 marks)*

(Total: 20 marks)

QUESTION FOUR

(a) Explain FIVE safe computing practices an everyday computer user should follow. *(10 marks)*

(b) Outline a logical troubleshooting approach a user should follow when a peripheral device stops working. *(10 marks)*

(Total: 20 marks)

QUESTION FIVE

(a) Explain what "phishing" is, and how a user can reduce the risk of falling victim to it. *(8 marks)*

(b) Explain TWO signs that a website may be safe to enter sensitive information into. *(6 marks)*

(c) State THREE ergonomic practices that reduce strain when using a computer for long periods. *(6 marks)*

(Total: 20 marks)

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13. Marking guide — model answers to the practice examination

QUESTION ONE

(a) *(6 marks)* — A computer system is the combination of hardware, software, and data working together to perform useful tasks. Its three main components: hardware (the physical machine), software (instructions telling the hardware what to do), and data (the information being processed).

(b) *(8 marks — 2 marks each, any four)* — System unit/case (houses core components); monitor (visual output); keyboard/mouse (input — typing/pointing); speakers (audio output); webcam/microphone (video/audio input).

(c) *(6 marks)* — An operating system manages the computer's hardware and provides the environment other software runs in (e.g. Windows). Productivity software is used for specific everyday tasks such as document creation (e.g. a word processor).

QUESTION TWO

(a) *(9 marks — 3 marks each, any three)* — Use clear, descriptive file names; organise files into a sensible folder structure by topic/date; avoid scattered duplicate copies; keep regular backups of important files.

(b) *(6 marks)* — An HDD uses traditional magnetic storage with moving parts, generally offering large capacity at lower cost. An SSD uses flash memory with no moving parts, generally much faster than an HDD.

(c) *(5 marks)* — Byte, kilobyte, megabyte, gigabyte, terabyte.

QUESTION THREE

(a) *(6 marks)* — Total = 32 GB = 32,000 MB. Used = 5 GB = 5,000 MB. Free = 32,000 − 5,000 = **27,000 MB**.

(b) *(8 marks)* — 27,000 MB ÷ 250 MB = **108 files**.

(c) *(6 marks — any two)* — External hard drive; optical media (CD/DVD); memory card.

QUESTION FOUR

(a) *(10 marks — 2 marks each, any five)* — Use strong, unique passwords with multi-factor authentication where available; keep the OS/applications patched; run an updated antivirus program; avoid opening attachments/links from unknown senders; keep regular backups stored separately; lock/log off the computer when leaving it unattended in a shared space.

(b) *(10 marks)* — Restart the device to clear minor glitches; check physical connections/cables; check for and install available updates/drivers; free up storage space if the drive is nearly full; read the exact error message for clues; escalate to qualified support if simple steps do not resolve it.

QUESTION FIVE

(a) *(8 marks)* — Phishing is a message or fake website designed to trick a user into revealing passwords or personal/financial information. Risk is reduced by not clicking links or opening attachments from unknown/suspicious senders, and by verifying a request independently before acting on it.

(b) *(6 marks — any two)* — A valid padlock/HTTPS indicator in the browser address bar; a familiar, correctly spelled domain name; the presence of legitimate contact/business information on the site.

(c) *(6 marks — any three)* — Position the monitor at roughly eye level and an arm's length away; sit with feet flat and good back support; take regular short breaks from the screen; keep the workspace well-lit and comfortable.

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14. Exam technique and revision checklist

- Storage-calculation questions: always convert every value to the same unit first (e.g. everything to MB) before doing the arithmetic — mixing units mid-calculation is the most common source of error.
- Hardware/software distinction questions: always give a concrete example alongside the definition — a definition alone often earns partial marks only.
- Safe-computing and troubleshooting questions: answer as a clear ordered or bulleted list, not one dense paragraph — this makes it easy to award a mark per correct point.

- Always read exactly how many items a question asks for ("state THREE," "explain FIVE") and provide exactly that many clearly separated points — extra unrequested points don't earn extra marks, and too few loses them.
- Keep definitions short, accurate, and in your own words; a precise one-line definition scores better than a long, vague paragraph.

Quick revision — one line per topic:

1. A computer system = hardware + software + data working together.
2. Core physical parts: system unit, motherboard, monitor, keyboard/mouse, speakers.
3. Software categories: operating system, productivity software, browsers, communication software, utilities.
4. Good file management: clear names, sensible folders, no scattered duplicates, regular backups.
5. HDDs are cheaper/larger but slower; SSDs are faster with no moving parts.
6. Storage units, smallest to largest: byte, kilobyte, megabyte, gigabyte, terabyte.
7. Common ports: USB (general peripherals), HDMI (video/audio out), Ethernet (wired network), audio jack.
8. Safe computing: strong passwords/MFA, updated software, antivirus, caution with links/attachments, regular backups.
9. Basic troubleshooting order: restart → check connections → check updates → free up space → read the error → escalate.
10. Ergonomics: eye-level monitor, good posture/support, regular breaks, clean/ventilated workspace.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.